1. Identification

Product identifier used on the label

Lupro-Cid®

Recommended use of the chemical and restriction on use

Recommended use*: feed additive(s)
Unsuitable for use: Not intended for sale to or use by the general public.

* The “Recommended use” identified for this product is provided solely to comply with a Federal requirement and is not part of the seller's published specification. The terms of this Safety Data Sheet (SDS) do not create or infer any warranty, express or implied, including by incorporation into or reference in the seller's sales agreement.

Details of the supplier of the safety data sheet

Company:
BASF Canada Inc.
5025 Creekbank Road
Building A, Floor 2
Mississauga, ON, L4W 0B6, CANADA

Telephone: +1 289 360-1300

Emergency telephone number

24 Hour Emergency Response Information
CHEMTREC: 1-800-424-9300
BASF HOTLINE: (800) 454-COPE (2673)

Other means of identification

Synonyms: Preparation based on: Formic acid, propionic acid, Water

2. Hazards Identification

According to Hazardous Products Regulations (HPR) (SOR/2015-17)

Classification of the product

Flam. Liq.	3	Flammable liquids
Acute Tox.	4 (Inhalation - vapour)	Acute toxicity
Acute Tox.	4 (oral)	Acute toxicity

Skin Corr./Irrit.	1B	Skin corrosion/irritation
Eye Dam./Irrit.	1	Serious eye damage/eye irritation
STOT SE	3 (irritating to respiratory system)	Specific target organ toxicity — single exposure

Label elements

Pictogram:

Signal Word:
Danger

Hazard Statement:

H226	Flammable liquid and vapour.
H314	Causes severe skin burns and eye damage.
H335	May cause respiratory irritation.
H302 + H332	Harmful if swallowed or if inhaled

Precautionary Statements (Prevention):

P271	Use only outdoors or in a well-ventilated area.
P280	Wear protective gloves, protective clothing and eye protection or face protection.
P210	Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
P260	Do not breathe dust/gas/mist/vapours.
P243	Take action to prevent static discharges.
P241	Use explosion-proof electrical, ventilating and lighting equipment.
P270	Do not eat, drink or smoke when using this product.
P264	Wash contaminated body parts thoroughly after handling.
P240	Ground and bond container and receiving equipment.
P242	Use only non-sparking tools.

Precautionary Statements (Response):

P310	Immediately call a POISON CENTER or physician.
P305 + P351 + P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P304 + P340	IF INHALED: Remove person to fresh air and keep comfortable for breathing.
P303 + P361 + P353	IF ON SKIN (or hair): Remove or Take off immediately all contaminated clothing. Rinse skin with water or shower.
P301 + P330 + P331	IF SWALLOWED: rinse mouth. Do NOT induce vomiting.
P363	Wash contaminated clothing before reuse.
P370 + P378	In case of fire: Use alcohol-resistant foam, carbon dioxide, dry powder or water spray for extinction.

Precautionary Statements (Storage):

P233	Keep container tightly closed.
P403 + P235	Store in a well-ventilated place. Keep cool.
P405	Store locked up.

Precautionary Statements (Disposal):

P501	Dispose of contents/container in accordance with local regulations.

Hazards not otherwise classified

No applicable information available.

Labeling of special preparations (GHS):
The following percentage of the mixture consists of components(s) with unknown hazards regarding the acute toxicity: 11 %

3. Composition / Information on Ingredients

According to Hazardous Products Regulations (HPR) (SOR/2015-17)

Formic Acid
CAS Number: 64-18-6
Content (W/W): 50.0 - 75.0%
Synonym: No data available.

propionic acid
CAS Number: 79-09-4
Content (W/W): 25.0 - 50.0%
Synonym: Propanoic acid; Propionic acid

4. First-Aid Measures

Description of first aid measures

General advice:
First aid personnel should pay attention to their own safety. If the patient is likely to become unconscious, place and transport in stable sideways position (recovery position). Immediately remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention. Immediately administer a corticosteroid from a controlled/metered dose inhaler.

If on skin:
Remove contaminated clothing. Wash skin with soap and water, rinse abundantly. Seek medical attention.

If in eyes:
Hold eyes open and rinse slowly and gently with water for 15 to 20 minutes. Remove contact lenses, if present, after first 5 minutes, then continue rinsing. If irritation develops, seek medical attention.

If swallowed:
Do not induce vomiting. Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

Most important symptoms and effects, both acute and delayed

Information on: Formic Acid
Symptoms: Overexposure may cause:, vomiting, aspiration pneumonia, circulatory collapse, death, acidosis, abdominal cramps, dyspnea, hypotension (low blood pressure), nausea, diarrhea, salivation
---

Indication of any immediate medical attention and special treatment needed

Note to physician

Treatment:	Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures

Suitable extinguishing media:
water spray, dry powder, alcohol-resistant foam, carbon dioxide

Unsuitable extinguishing media for safety reasons:
water jet

Special hazards arising from the substance or mixture
Hazards during fire-fighting:
harmful vapours, carbon oxides
The substances/groups of substances mentioned can be released in case of fire. The product is combustible.

Advice for fire-fighters
Protective equipment for fire-fighting:
Wear self-contained breathing apparatus and chemical-protective clothing.

Further information:
Dispose of fire debris and contaminated extinguishing water in accordance with official regulations.
Cool endangered containers with water-spray.

6. Accidental release measures

Personal precautions, protective equipment and emergency procedures
Use personal protective clothing. Information regarding personal protective measures, see section 8. Ensure adequate ventilation. Do not breathe vapour/spray. Wear respiratory protection if ventilation is inadequate. Avoid contact with the skin, eyes and clothing. Take off immediately all contaminated clothing. Avoid all sources of ignition: heat, sparks, open flame.

Environmental precautions
Do not discharge into drains/surface waters/groundwater.

Methods and material for containment and cleaning up
For small amounts: Contain with absorbent material (e.g. sand, silica gel, acid binder, general purpose binder, sawdust).
For large amounts: Dike spillage. Cover with blanket of foam (alcohol-resistant foam). Pump off product.
For residues: Pick up with suitable absorbent material.

Dispose of absorbed material in accordance with regulations. Cleaning operations should be carried out only while wearing breathing apparatus.

7. Handling and Storage

Precautions for safe handling

Ensure thorough ventilation of stores and work areas. Wear suitable protective clothing and eye/face protection. Avoid contact with the skin, eyes and clothing. Keep container tightly sealed.

Protection against fire and explosion:
The product is combustible. Avoid all sources of ignition: heat, sparks, open flame. Take precautionary measures against static discharges. If exposed to fire, keep containers cool by spraying with water. Vapours may form explosive mixture with air.

Conditions for safe storage, including any incompatibilities

Segregate from alkalies and alkalizing substances.

Suitable materials for containers: glass, Stainless steel 1.4401, Stainless steel 1.4301 (V2), Aluminium, High density polyethylene (HDPE), Low density polyethylene (LDPE)

Further information on storage conditions: Keep container tightly closed and dry; store in a cool place. Protect containers from physical damage.

8. Exposure Controls/Personal Protection

Components with occupational exposure limits

Formic Acid	ACGIH, US:	TWA value 5 ppm ;
	ACGIH, US:	STEL value 10 ppm ;
	OSHA Z1:	PEL 5 ppm 9 mg/m3 ;
propionic acid	ACGIH, US:	TWA value 10 ppm ;

Personal protective equipment

Respiratory protection:
Respiratory protection in case of vapour/aerosol release. Wear a NIOSH-certified (or equivalent) respirator as necessary.

Hand protection:
Wear impermeable chemical resistant protective gloves.

Eye protection:
Wear face shield or tightly fitting safety goggles (chemical goggles) if splashing hazard exists.

Body protection:
Body protection must be chosen based on level of activity and exposure.

General safety and hygiene measures:
Avoid contact with the skin, eyes and clothing. Do not inhale gases/vapours/aerosols. Wearing of closed work clothing is required additionally to the stated personal protection equipment. No eating, drinking, smoking or tobacco use at the place of work. Hands and/or face should be washed before breaks and at the end of the shift. Store work clothing separately.

9. Physical and Chemical Properties

Form:	liquid	
Odour:	pungent	
Odour threshold:	Not determined due to potential health hazard by inhalation.	
Colour:	Colourless to reddish	
pH value:	1.5 - 1.9 (100 g/l)	
Melting point:	-18.2 °C	
Freezing point:	No data available.	
Boiling range:	107 - 117 °C	(DIN 53171)
Flash point:	55.5 °C	(DIN 51755)
Flammability:	Flammable liquid and vapour.	(derived from flash - and boiling point)
Lower explosion limit:	For liquids not relevant for classification and labelling. The lower explosion point may be 5 - 15 °C below the flash point.	
Upper explosion limit:	For liquids not relevant for classification and labelling.	
Autoignition:	515 °C	
Vapour pressure:	24 hPa (20 °C)	
Density:	approx. 1.15 g/cm3 (20 °C)	
Vapour density:	> 1 (20 °C) Heavier than air.	(calculated)
Partitioning coefficient n-octanol/water (log Pow):	not applicable for mixtures	
Information on: Formic Acid		
Partitioning coefficient n-octanol/water (log Pow):	-2.1 (23 °C) -1.9 (23 °C) -2.3 (23 °C)	(Directive 92/69/EEC, A.8) (Directive 92/69/EEC, A.8) (Directive 92/69/EEC, A.8)
Information on: propionic acid		
Partitioning coefficient n-octanol/water (log Pow):	0.33 Literature data.	(measured)

Thermal decomposition:	No data available.
Information on: Formic Acid	
Thermal decomposition:	*350 °C, > 150 kJ/kg (DSC (DIN 51007))*
Information on: propionic acid	
Thermal decomposition:	*(DSC (DIN 51007))* *No exothermic decomposition within the mentioned temperature range. It is not a self-decompositionable substance.*

Viscosity, dynamic:	approx. 1.86 mPa.s (20 °C)	
Viscosity, kinematic:	1.61 mm2/s (20 °C) calculated	(calculated (from dynamic viscosity))
Particle size:	The substance / product is marketed or used in a non solid or granular form.	
Solubility in water:	fully soluble	
Molar mass:	No data available.	

10. Stability and Reactivity

Reactivity
No hazardous reactions if stored and handled as prescribed/indicated.

Corrosion to metals:
In the presence of water or moisture metal corrosion cannot be excluded. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Oxidizing properties:
Based on its structural properties the product is not classified as oxidizing.

Chemical stability
The product is stable if stored and handled as prescribed/indicated.

Possibility of hazardous reactions
Reacts with alkalies. Exothermic reaction.

Conditions to avoid
Avoid electro-static discharge. Avoid all sources of ignition: heat, sparks, open flame.

Incompatible materials
alkalies

Hazardous decomposition products

Decomposition products:
Hazardous decomposition products: No hazardous decomposition products if stored and handled as prescribed/indicated.

Thermal decomposition:
No data available.

11. Toxicological information

Primary routes of exposure

Routes of entry for solids and liquids are ingestion and inhalation, but may include eye or skin contact. Routes of entry for gases include inhalation and eye contact. Skin contact may be a route of entry for liquefied gases.

Acute Toxicity/Effects

Acute toxicity
Assessment of acute toxicity: Of moderate toxicity after short-term inhalation. Of moderate toxicity after single ingestion.

Oral

Information on: Formic Acid
Type of value: LD50
Species: rat (male/female)
Value: 730 mg/kg (OECD Guideline 401)

Information on: propionic acid
Type of value: LD50
Species: rat (male/female)
Value: 3,455 mg/kg (similar to OECD guideline 401)

Inhalation

Information on: Formic Acid
Type of value: LC50
Species: rat (male/female)
Value: 7.85 mg/l (BASF-Test)
Exposure time: 4 h
The vapour was tested.

Information on: propionic acid
Type of value: LC50
Species: rat (male/female)
Value: > 19.7 mg/l (OECD Guideline 403)
Exposure time: 1 h
The vapour was tested.

Type of value: LC0
Species: rat (male/female)
Value: 24.4 mg/l (IRT)
Exposure time: 8 h
The vapour was tested.
Literature data. No mortality within the stated exposition time as shown in animal studies.

Dermal

Information on: propionic acid
Type of value: LD50
Species: rat (female)
Value: 3,235 mg/kg (similar to OECD guideline 402)

Assessment other acute effects
Assessment of STOT single:
Causes temporary irritation of the respiratory tract.

Irritation / corrosion

Assessment of irritating effects: Corrosive! Damages skin and eyes. May cause severe damage to the eyes.

Information on: Formic Acid
Assessment of irritating effects: Highly corrosive! Damages skin and eyes.

Information on: propionic acid
Assessment of irritating effects: Corrosive! Damages skin and eyes.

Sensitization
Assessment of sensitization: Skin sensitizing effects were not observed in animal studies.

Information on: Formic Acid
Buehler test
Species: guinea pig
Result: Non-sensitizing.
Method: OECD Guideline 406

Information on: propionic acid
Guinea pig maximization test
Species: guinea pig
Result: Non-sensitizing.
Method: similar to OECD guideline 406
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Aspiration Hazard
No aspiration hazard expected.

Chronic Toxicity/Effects

Repeated dose toxicity
Assessment of repeated dose toxicity: Based on available data, the classification criteria are not met.

Information on: Formic Acid
Assessment of repeated dose toxicity: No substance-specific organtoxicity was observed after repeated administration to animals. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Information on: propionic acid
Assessment of repeated dose toxicity: No substance-specific organtoxicity was observed after repeated administration to animals. After repeated administration the prominent effect is the induction of corrosion.

Genetic toxicity
Assessment of mutagenicity: No mutagenic effect was found in various tests with bacteria and mammalian cell culture.

Information on: Formic Acid
Assessment of mutagenicity: No mutagenic effect was found in various tests with bacteria and mammalian cell culture. The substance was not mutagenic in an insect test.

Information on: propionic acid
Assessment of mutagenicity: The substance was not mutagenic in bacteria. The substance was not mutagenic in mammalian cell culture. The substance was not mutagenic in studies with mammals. The product has not been fully tested. The statements have been derived in parts from products of a similar structure or composition.

Carcinogenicity
Assessment of carcinogenicity: In long-term animal studies in which the substance was given in high concentrations by feed, a carcinogenic effect was not observed.

Information on: Formic Acid
Assessment of carcinogenicity: In long-term studies in rats and mice in which the substance was given by feed, a carcinogenic effect was not observed. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Information on: propionic acid
Assessment of carcinogenicity: In long-term animal studies in which the substance was given in high concentrations by feed, a carcinogenic effect was not observed.

Reproductive toxicity
Assessment of reproduction toxicity: The results of animal studies gave no indication of a fertility impairing effect.

Information on: Formic Acid
Assessment of reproduction toxicity: The results of animal studies gave no indication of a fertility impairing effect. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Information on: propionic acid
Assessment of reproduction toxicity: No data available. Study scientifically not justified.

Teratogenicity
Assessment of teratogenicity: No indications of a developmental toxic / teratogenic effect were seen in animal studies.

Information on: Formic Acid
Assessment of teratogenicity: No indications of a developmental toxic / teratogenic effect were seen in animal studies. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Information on: propionic acid
Assessment of teratogenicity: No indications of a developmental toxic / teratogenic effect were seen in animal studies. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Other Information
The product has not been tested. The statement has been derived from the properties of the individual components.

12. Ecological Information

Toxicity

Aquatic toxicity
Assessment of aquatic toxicity:
There is a high probability that the product is not acutely harmful to aquatic organisms. The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations.
The product gives rise to pH shifts.

Assessment of terrestrial toxicity
No data available.

Toxicity to terrestrial plants

Information on: propionic acid
EC50 (3 d) 125.8 mg/l 188.7 mg/kg, Lactuca sativa
Literature data.

Microorganisms/Effect on activated sludge

Toxicity to microorganisms

Information on: Formic Acid
other aerobic
activated sludge, domestic, non-adapted/EC10 (13 d): 72 mg/l

Information on: propionic acid
DIN EN ISO 8192 aquatic
activated sludge, domestic/EC20 (30 min): 500 - 1,040 mg/l
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Persistence and degradability

Assessment biodegradation and elimination (H2O)
Readily biodegradable (according to OECD criteria).

Bioaccumulative potential

Assessment bioaccumulation potential
Significant accumulation in organisms is not to be expected.

Mobility in soil

Assessment transport between environmental compartments
The substance will not evaporate into the atmosphere from the water surface.
Adsorption to solid soil phase is not expected.

Additional information

Add. remarks environm. fate & pathway:
The product has not been tested. The statements on environmental fate and pathway have been derived from the properties of the individual components.

Other ecotoxicological advice:
The product has not been tested. The statement has been derived from the properties of the individual components.

13. Disposal considerations

Waste disposal of substance:
Observe national and local legal requirements.

Container disposal:
Dispose of in accordance with national, state and local regulations.

14. Transport Information

Land transport
TDG

Hazard class:	8
Packing group:	II
ID number:	UN 2920
Hazard label:	8, 3
Proper shipping name:	CORROSIVE LIQUID, FLAMMABLE, N.O.S. (contains FORMIC ACID, PROPIONIC ACID)

Sea transport
IMDG

Hazard class:	8
Packing group:	II
ID number:	UN 2920
Hazard label:	8, 3
Marine pollutant:	NO
Proper shipping name:	CORROSIVE LIQUID, FLAMMABLE, N.O.S. (contains FORMIC ACID, PROPIONIC ACID)

Air transport
IATA/ICAO

Hazard class:	8
Packing group:	II
ID number:	UN 2920
Hazard label:	8, 3
Proper shipping name:	CORROSIVE LIQUID, FLAMMABLE, N.O.S. (contains FORMIC ACID, PROPIONIC ACID)

15. Regulatory Information

Federal Regulations

Registration status:

Feed	DSL, CA	released / listed
Chemical	DSL, CA	released / listed

NFPA Hazard codes:
Health: 3 Fire: 2 Reactivity: 0 Special:

Assessment of the hazard classes according to UN GHS criteria (most recent version):

Flam. Liq.	3	Flammable liquids
Acute Tox.	4 (oral)	Acute toxicity
Acute Tox.	4 (Inhalation - vapour)	Acute toxicity
Acute Tox.	5 (dermal)	Acute toxicity
Skin Corr./Irrit.	1B	Skin corrosion/irritation
Eye Dam./Irrit.	1	Serious eye damage/eye irritation
STOT SE	3 (irritating to respiratory system)	Specific target organ toxicity — single exposure

16. Other Information

SDS Prepared by:
BASF NA Product Regulations
SDS Prepared on: 2022/11/03

We support worldwide Responsible Care® initiatives. We value the health and safety of our employees, customers, suppliers and neighbors, and the protection of the environment. Our commitment to Responsible Care is integral to conducting our business and operating our facilities in a safe and environmentally responsible fashion, supporting our customers and suppliers in ensuring the safe and environmentally sound handling of our products, and minimizing the impact of our operations on society and the environment during production, storage, transport, use and disposal of our products.

Lupro-Cid® is a registered trademark of BASF Canada or BASF SE
END OF DATA SHEET